



005 Week 1 — English Worksheet

Topic: One Function, Many Coasters — Parameters · **Course:** Theme Park Creator · **Time:** about 45 minutes

This week you build a roller coaster with a **function**. You give the function **parameters** — numbers it uses inside — so the same function can build a long coaster, a short coaster, a tall spiral, or a wide one, just by changing the numbers you **call** it with.

These are the blocks you use this week:

```
rollerCoasterBuilder.add_straight_line(10)
rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, 10)
rollerCoasterBuilder.add_spiral(RcbVerticalDirection.UP, TurnDirection.LEFT, 10, 3)
rollerCoasterBuilder.add_turn(TurnDirection.LEFT)
```

`add_spiral(...)` 's last two numbers are the **height** then the **width** of the spiral. A direction is `UP` or `DOWN`. A turn is `LEFT` or `RIGHT`.

1 · Predict 🧠

Read each function. Before you imagine the cart riding it, write what you think will happen.

```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)

build_hill(8)
```

The parameter `line_len` is used twice. How long is the straight part, and how tall is the ramp?


```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)

build_hill(4)
build_hill(12)
```

The same function is called twice with different numbers. Are the two hills the same size?
Which one is bigger?


```
def build_spiral(spiral_height, spiral_width):
    rollerCoasterBuilder.add_spiral(RcbVerticalDirection.UP, TurnDirection.LEFT,
    spiral_height, spiral_width)

build_spiral(10, 3)
```

This function has two parameters. Which number sets how *tall* the spiral is, and which sets how *wide*?

2 · Spot the Bug 🐛

Each block of code was meant to do something, but it is broken. Read what it is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The code should build a hill 8 long. But when you run it, **nothing happens at all**.

```
def build_hill(line_len):  
    rollerCoasterBuilder.add_straight_line(line_len)  
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)
```

Hint: writing `def` only **teaches** the function. Something is missing after it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The function should build a hill as long as the number you give it. But `build_hill(20)` and `build_hill(3)` build the **same** hill every time.

```
def build_hill(line_len):  
    rollerCoasterBuilder.add_straight_line(10)  
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, 10)  
  
build_hill(20)
```

Hint: look at the numbers inside. The function never actually uses `line_len`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This coaster should go **up** and then come back **down** so the cart returns to ground level. But it just keeps climbing into the sky and never comes down.

```
def up_and_down(ramp_len):  
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, ramp_len)  
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, ramp_len)  
  
up_and_down(10)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Build the Loop Coaster

Now the real challenge. In your homework world, write **one** function called `loop_coaster` that builds a coaster which **loops all the way back to the starting area**. Your function must have these **four parameters**:

```
def loop_coaster(line_len, ramp_len, spiral_height, spiral_width):
```

Your coaster must include, in total:

- **2 ramps** — one going **UP**, one coming back **DOWN**
- **2 spirals** — one going **UP**, one coming back **DOWN**
- enough **turns** that the track curves around and ends back where it started

Here are the building blocks you need. Right now they all use **fixed numbers** and all point the **same way** — your job is to fix that:

```
rollerCoasterBuilder.add_straight_line(10)
rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, 10)
rollerCoasterBuilder.add_spiral(RcbVerticalDirection.UP, TurnDirection.LEFT, 10, 3)
rollerCoasterBuilder.add_turn(TurnDirection.LEFT)
```

To finish the challenge you must:

1. **Swap the numbers for parameters** — replace `10`, `10`, `10`, `3` with `line_len`, `ramp_len`, `spiral_height`, `spiral_width` so the size is controlled by the call.
2. **Flip the directions on the way back** — change the second ramp and the second spiral from `UP` to `DOWN` so the cart returns to ground level.
3. **Add turns** — keep the `add_turn(TurnDirection.LEFT)` blocks coming until the track curves all the way around and meets the start. Test in the world, then add or remove a turn until it closes the loop.

Finally, **call your function once** with numbers you choose, for example:

```
loop_coaster(8, 10, 10, 3)
```

loop_coaster

4 · Tell Me What You Built 📷

When your coaster loops back to the start, ride it once in your world. Then send a photo or video and explain what you did. Use these sentence starters — write 4 to 6 sentences total.

My four parameters were ...

The number I changed that made the biggest difference was ...

To make the cart come back down, I ...

To make the track loop back to the start, I ...

One tricky moment was when ...

If I had more time, I would ...

5 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) of your code and your coaster. Talk through it like you are teaching someone who has never seen it. Try to use these words: **function**, **parameter**, **call**, **UP / DOWN**, **turn**.

1. Show your `loop_coaster` function and read the line with the four **parameters** out loud.
2. Point at one parameter and say what it controls.
3. Run your **call** and show the cart riding the loop all the way back to the start.
4. Say in your own words what a **parameter** is and why one function can build many coasters.

Write what you will say in your video. Plan it here before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.